Notes: Leland and Pyle (JF, 1977)

Informational Asymmetries, Financial Structure, and Financial Intermediation

Contents

1 Big picture

- **Question.** How does asymmetric information about project quality affect financing, firm value, capital structure, and the existence of financial intermediation?
- **Setting.** An entrepreneur knows the expected return of his own project, but outside investors do not. The entrepreneur can retain a fraction α of the project's equity, sell the rest to the market, and trade the market portfolio and a riskless asset.
- **Core challenge.** If lenders cannot tell good projects from bad ones, market prices reflect average quality. That can shut down financing for good projects because low-quality entrepreneurs would like to sell overpriced claims to uninformed investors.
- **Main signaling idea.** The entrepreneur's willingness to keep equity in the project can reveal information. "Actions speak louder than words": retaining more equity is costly, so it can be a credible signal of project quality.
- **Main theoretical result.** In equilibrium, under normal asset demand, the market valuation schedule $\mu(\alpha)$ is increasing in insider retention α. Better projects are financed by entrepreneurs who keep larger ownership stakes.
- **Capital-structure result.** The value of the firm rises with insider ownership, and debt is generally related to value even without taxes. This is a sharp contrast with the Modigliani-Miller benchmark.
- **Risk result.** Firms with riskier returns use less debt even when bankruptcy costs are absent. The mechanism is informational, not bankruptcy-based.
- **Efficiency result.** Signaling is costly because entrepreneurs hold too much equity in their own firms relative to the full-information benchmark. But the set of projects undertaken is the same as under costless information transfer.
- **Intermediation result.** Financial intermediaries can be understood as institutions that gather private information, hold assets themselves, and signal information quality through insider capital at risk.
- **Economic takeaway.** Ownership structure is not just financing. It is also information transmission.

2 Setup and economic objects (key objects)

2.1 Project, entrepreneur, and information structure

- **Project return.** The investment project requires a capital outlay K and yields future return $\mu + \tilde{x}$, where μ is the mean return known to the entrepreneur and $\tilde{x}$ is a zero-mean random variable with variance σ_x^2.
- **Private information.** The entrepreneur knows the true μ of his own project. Outside investors only have a subjective distribution over μ.
- **Why cheap talk fails.** The entrepreneur cannot credibly reveal μ directly because exaggeration is privately attractive and verification is costly or impossible.

- **Observable signal.** The market can observe α, the fraction of equity retained by the entrepreneur. The paper treats α as the key signal of type.

2.2 Financial instruments and portfolio choices

- **Debt.** The firm can issue riskless debt D that promises $(1 + r)D$ next period.
- **Market portfolio.** $\tilde{M}$ is the gross return on the market portfolio, V_M its current value, and β the amount of the market portfolio held by the entrepreneur.
- **Riskless asset.** Y denotes the entrepreneur's personal holdings of the riskless asset.
- **Important assumption.** Both the firm and the entrepreneur can borrow or lend at the same riskless rate r. Later, the paper introduces small transaction-cost or institutional restrictions to discuss debt more meaningfully.

2.3 Value schedule and equilibrium concept

Given signal α, the market prices the project using a valuation schedule $\mu(\alpha)$:

$$V(\alpha) = \frac{1}{1+r}\left[\mu(\alpha) - \lambda\right], \tag{1}$$

where λ is the market's adjustment for project risk.

- **Interpretation.** $V(\alpha)$ is not just discounted expected value. It is the discounted *market-perceived* expected return net of a risk adjustment.
- **Equilibrium requirement.** A valuation schedule is an equilibrium schedule if the market correctly infers the entrepreneur's type from his chosen ownership share:

$$\mu[\alpha^*(\mu)] = \mu \quad \text{for all } \mu \text{ for which the project is undertaken.} \tag{5}$$

- **Threshold for investment.** The entrepreneur compares project financing to his reservation utility from not undertaking the project. This defines a cutoff μ^*: only projects with sufficiently high μ are financed.

2.4 Assumptions doing the main work

- **Competitive capital markets.** Prices must satisfy zero-profit logic for outside investors.
- **Small project assumption.** The entrepreneur treats the project as too small to affect the return or price of the market portfolio.
- **Differentiability.** The schedule $\mu(\alpha)$ is assumed differentiable.
- **Normal asset demand.** This assumption is crucial for monotone signaling. It means that in a standard portfolio problem, demand for an asset rises when its price falls.
- **Existence is assumed, not solved in general.** The paper characterizes properties of equilibrium schedules and then computes one explicitly in a tractable example.

3 Model and equilibrium characterization (all equations + interpretation)

3.1 Budget constraint and end-of-period wealth

If the entrepreneur undertakes the project, his choices satisfy

$$W_0 + D + (1-\alpha)\big[V(\alpha) - D\big] - K - \beta V_M - Y = 0. \tag{2}$$

End-of-period wealth is

$$\tilde{W}_1 = \alpha\big[\tilde{x} + \mu - (1+r)D\big] + \beta\tilde{M} + (1+r)Y.$$

Substituting the budget constraint and the valuation formula gives

$$\tilde{W}_1 = \alpha\big[\tilde{x} + \mu - \mu(\alpha) + \lambda\big] + \beta\big[\tilde{M} - (1+r)V_M\big] + (W_0 - K)(1+r) + \mu(\alpha) - \lambda. \tag{3}$$

The entrepreneur solves

$$\max_{\alpha,\beta}\ \mathbb{E}\Big[U(\tilde{W}_1)\Big]. \tag{4}$$

Interpretation.

- The entrepreneur chooses how much project risk to keep, how much market risk to hold, and how much to borrow or lend overall.
- A key subtlety is that D disappears from the utility maximization problem once wealth is written in reduced form. Without transaction costs, only the *net* borrowing position matters, not whether borrowing happens inside the firm or personally.
- This is why the paper delays the debt discussion until later. At this stage, the core signal is insider equity retention α, not leverage D.

3.2 First-order conditions and the signaling wedge

For any equilibrium schedule $\mu(\alpha)$, optimality requires

$$\frac{\partial\mathbb{E}[U(\tilde{W}_1)]}{\partial\alpha} = \mathbb{E}\Big[U'(\tilde{W}_1)\,(\tilde{x} + \mu - \mu(\alpha) + \lambda + (1-\alpha)\mu_\alpha)\Big] = 0, \tag{6}$$

$$\frac{\partial\mathbb{E}[U(\tilde{W}_1)]}{\partial\beta} = \mathbb{E}\Big[U'(\tilde{W}_1)\,\Big(\tilde{M} - (1+r)V_M\Big)\Big] = 0, \tag{7}$$

where $\mu_\alpha \equiv d\mu(\alpha)/d\alpha$.

Using the equilibrium condition $\mu[\alpha^*(\mu)] = \mu$, equation (6) becomes

$$(1-\alpha)\mu_\alpha = -\frac{\mathbb{E}\Big[U'(\tilde{W}_1)(\tilde{x}+\lambda)\Big]}{\mathbb{E}\Big[U'(\tilde{W}_1)\Big]}. \tag{8}$$

Second-order conditions are summarized by

$$A < 0, \qquad B < 0, \qquad AB - C^2 > 0, \tag{9}$$

for appropriately defined Hessian terms A, B, C.

Interpretation.

- In an ordinary portfolio problem, the "price" of the risky asset is fixed. Here it is not: the market price $V(\alpha)$ depends on the entrepreneur's own choice of α because outsiders read α as a signal.
- Equation (8) is the heart of the model. The left side is the marginal gain in market valuation from retaining more equity. The right side is the marginal utility cost of bearing extra project risk.
- Signaling is possible because higher-type entrepreneurs are more willing to pay this cost.

3.3 Monotonicity and the welfare cost of signaling

Normal Asset Demand. The paper defines an asset as normal if, in a standard portfolio problem without signaling, demand for that asset rises when its price falls.

Theorem I. The equilibrium valuation schedule $\mu(\alpha)$ is strictly increasing in α over the relevant domain if and only if the entrepreneur's demand for equity in his project is normal.

Interpretation.

- Better projects induce entrepreneurs to hold more equity.
- The market therefore rationally interprets higher insider retention as good news.
- This is the basic separating mechanism.

Theorem II. In equilibrium with signaling by α, entrepreneurs with normal demands hold more equity in their own projects than they would if they could costlessly communicate μ.

Interpretation.

- Signaling is costly because entrepreneurs distort their portfolios away from the full-information optimum.
- That distortion is the welfare cost of signaling.
- But the cost is not pure waste in the sense of shutting down financing. The signal may be exactly what allows good projects to be financed at all.

3.4 Tractable example: mean-variance utility and CAPM pricing

The paper then studies a special case in which

$$\mathbb{E}\Big[U(\tilde{W}_1)\Big] = G\Big[\mathbb{E}(\tilde{W}_1) - \frac{b}{2}\sigma^2(\tilde{W}_1)\Big], \tag{10}$$

where G is increasing and b is a risk-aversion parameter.

Risk adjustment is

$$\lambda = \lambda^* \mathrm{Cov}(\tilde{x}, \tilde{M}), \qquad \lambda^* = \frac{\mathbb{E}(\tilde{M}) - (1+r)V_M}{\sigma_M^2}. \tag{11}$$

The first-order conditions become

$$\big[\mu - \mu(\alpha) + \lambda^* \mathrm{Cov}(\tilde{x}, \tilde{M})\big] + (1-\alpha)\mu_\alpha - \alpha b \sigma_x^2 - \beta b \mathrm{Cov}(\tilde{x}, \tilde{M}) = 0, \tag{12}$$

$$\big[\mathbb{E}(\tilde{M}) - (1+r)V_M\big] - \alpha b \mathrm{Cov}(\tilde{x}, \tilde{M}) - \beta b \sigma_M^2 = 0. \tag{13}$$

Combining these yields

$$(1-\alpha)\mu_\alpha = b\alpha Z, \tag{14}$$

where

$$Z = \frac{\sigma_x^2 \sigma_M^2 - \big[\mathrm{Cov}(\tilde{x}, \tilde{M})\big]^2}{\sigma_M^2}.$$

The solution family is

$$\mu(\alpha) = -bZ\big[\log(1-\alpha) + \alpha\big] + C. \tag{15}$$

Interpretation.

- Z is the project's *specific risk*. If project return is independent of the market, $Z = \sigma_x^2$. If it is perfectly correlated with the market, $Z = 0$.
- The schedule is steeper when either risk aversion b or specific risk Z is higher.
- Economically, low types dislike holding project-specific risk less in utility terms than high types dislike being pooled with low types. That difference sustains separation.

3.5 From a family of schedules to the unique competitive schedule

The family in equation (15) is too large. Competition among lenders selects one schedule.

- **Curves above and to the left of KK'.** These imply $\mu(0) > (1+r)K + \lambda$. Then even entrepreneurs with arbitrarily poor projects can retain zero equity and still obtain financing at terms that make them better off. This attracts "freeloaders" and violates equilibrium.
- **Curves below and to the right of KK'.** These can satisfy the inference requirement, but they impose unnecessarily high signaling costs. If some lenders offer KK' while others offer such schedules, all entrepreneurs prefer KK'.

So the unique equilibrium schedule is

$$\mu(\alpha) = -bZ\big[\log(1-\alpha) + \alpha\big] + (1+r)K + \lambda, \qquad \alpha > 0. \tag{16}$$

Using equation (1),

$$V(\alpha) = \frac{1}{1+r}\left[-bZ\big(\log(1-\alpha) + \alpha\big)\right] + K, \qquad V_\alpha = \frac{bZ}{1+r}\left[\frac{\alpha}{1-\alpha}\right] > 0, \tag{17}$$

so

$$V(0) = K, \qquad V(\alpha) > K \quad \text{for } \alpha > 0.$$

Interpretation.

- The equilibrium schedule starts exactly at the project's cost. This is what rules out freeloading by zero-retention low-quality types.

- Higher insider retention strictly raises firm value because it induces a more favorable market inference about μ.
- This is the paper's cleanest departure from a world in which financial structure is irrelevant.

3.6 Project selection, comparative statics, and welfare

Proposition I. A project is undertaken if and only if its true market value, given μ, exceeds its cost.

Interpretation.

- This is the paper's main efficiency result.
- The signaling distortion affects *how* projects are financed, not *which* positive-NPV projects get financed.
- Put differently: the intensive margin is distorted, but the extensive margin is efficient.

Proposition II. An increase in either specific risk Z or risk aversion b reduces the entrepreneur's equilibrium equity position $\alpha^*(\mu)$ for any project that is undertaken.

Interpretation.

- When signaling becomes more personally costly, entrepreneurs retain less equity for a given project quality.
- The schedule adjusts so that separation still works, but with lower retained ownership.

Proposition III. An increase in specific risk Z raises the entrepreneur's expected utility for any undertaken project.

Interpretation.

- This is initially counterintuitive.
- The logic is that higher *specific* risk makes the project more distinct from the market portfolio and therefore easier to signal.
- In equilibrium, that lowers the cost of separating good types from bad ones.

3.7 Debt, firm value, and the Modigliani-Miller theorem

The paper next asks how value V relates to the financing decision D.

From the budget constraint,

$$H \equiv \alpha D - Y = K - W_0 - (1-\alpha)V(\alpha) + \beta V_M. \tag{18}$$

In the example with project return independent of the market, and focusing on the region with positive debt, the paper sets $Y = 0$ and, for simplicity, $r = 0$. Then

$$D = \frac{K - W_0 + \beta V_M - (1-\alpha)V(\alpha)}{\alpha} = \frac{K - W_0 + \beta V_M - (1-\alpha)\left[-bZ\big(\log(1-\alpha)+\alpha\big) + K\right]}{\alpha}. \tag{19}$$

Differentiating with respect to α yields

$$\operatorname{sign}\left(\frac{\partial D}{\partial \alpha}\right) = \operatorname{sign}\Big\{-bZ\left[\log(1-\alpha)+\alpha+\alpha^2\right]+W_0-\beta V_M\Big\}. \tag{20}$$

If

$$\frac{W_0-\beta V_M}{K} \geq .186,$$

then $\partial D/\partial\alpha > 0$ whenever $D > 0$.

Proposition IV. For any level of μ, greater project variance σ_x^2 implies lower optimal debt.

Interpretation.

- The model predicts a negative relation between project risk and debt even without bankruptcy costs.
- That result comes purely from signaling and portfolio choice.

How this relates to Modigliani-Miller.

- Across observationally similar firms, both V and D are increasing in α and therefore in μ. So a regression of value on debt would show a positive relation.
- But this is *statistical*, not *causal.* Holding α fixed, a change in D does not change perceived returns, so capital structure remains irrelevant in the MM sense.
- The key point is that observed debt may be correlated with the informative choice α once transaction costs or institutional constraints connect them.

3.8 Financial intermediation

The final section of the paper explains financial intermediation as an informational response rather than a mere transactions-cost story.

- **Why direct information sale is hard.** A firm that collects information and tries to sell it directly faces two problems:
  - *Appropriability.* Information is partly a public good. Buyers may share or resell it.
  - *Credibility.* Buyers may not know whether the information is good or bad. A lemons problem appears in the market for information itself.
- **Why intermediation helps.** If the information-gathering firm buys and holds assets itself, the return to information is embodied in a private good: the intermediary's portfolio return.
- **Why insider capital matters.** The intermediary's own willingness to hold equity becomes a signal that it has genuinely good information.
- **Why intermediaries are leveraged.** If intermediary assets have low specific risk, Proposition IV implies high leverage (via debt or deposits), which matches a central empirical feature of financial intermediaries.
- **Sorting logic.** Once an intermediary is better than the market at screening a class of risks, the best borrowers have strong incentives to sort into that intermediary. That starts a "peeling off" process that can pull nearly all but the worst risks into the intermediary sector.

4 What makes the argument credible (and where it is limited)

4.1 Why equity retention is a credible signal

- The signal is costly. Retaining equity forces the entrepreneur to bear project-specific risk.
- Low-quality entrepreneurs do not want to mimic high-quality ones if doing so requires holding too much risky equity.
- This is the Spence-style self-selection logic embedded in a finance model.

4.2 What does the heavy lifting mathematically

- The equilibrium condition $\mu[\alpha^*(\mu)] = \mu$ ties beliefs to actions.
- Competitive pricing forces market valuations to be consistent with zero expected excess returns for outside investors.
- Normal asset demand delivers the monotonicity needed for a separating schedule.
- In the tractable example, the CAPM-style risk adjustment and mean-variance preferences turn the general problem into a solvable differential equation.

4.3 What the paper proves and what it assumes

- The paper does *not* solve the general existence problem for signaling equilibrium. It assumes existence in the general model and then explicitly constructs an equilibrium in the example.
- The signal is noiseless once observed: the market infers type exactly from α in equilibrium.
- The analysis is essentially static around a single financing decision, even though the paper speaks to broader financial structure questions.

4.4 Main limitations

- **Riskless debt assumption.** The baseline assumes both the firm and the entrepreneur can issue debt at the riskless rate. This is analytically convenient but restrictive.
- **One-dimensional private information.** Project quality is summarized by a single scalar μ.
- **Single signal.** The paper focuses on insider ownership and largely abstracts from other signals such as dividends, disclosure policy, collateral, or covenants.
- **Preliminary intermediation section.** The discussion of intermediaries is conceptual rather than a fully worked-out equilibrium model.

4.5 Relation to the literature

- The paper sits directly on the early asymmetric-information tradition of Akerlof, Spence, Rothschild-Stiglitz, and Riley.

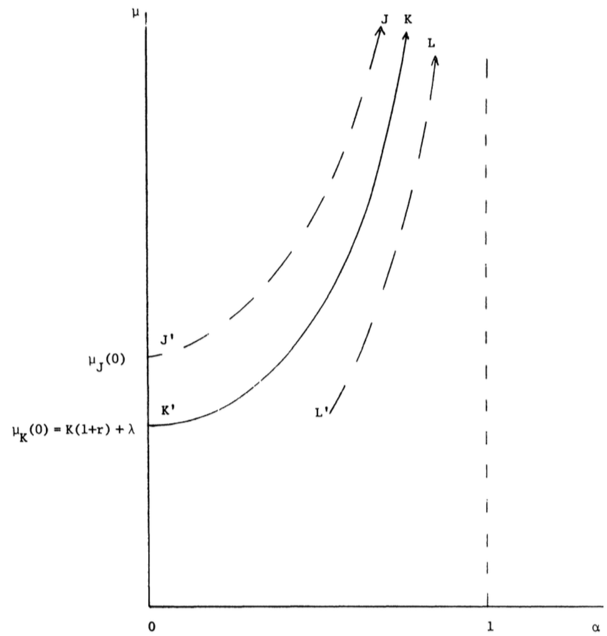

FIGURE 1. Equilibrium signaling schedules

- Relative to Modigliani-Miller, it shows how information asymmetry can make observed financing choices informative about firm value even without taxes.
- Relative to later corporate-finance signaling models, this paper is an early foundation for the idea that insider ownership, leverage, and financial claims can transmit private information.

5 Figure 1 and key propositions

5.1 Figure 1: equilibrium signaling schedules

Read:

- Figure 1 draws a family of candidate signaling schedules of the form

$$\mu(\alpha) = -bZ[\log(1-\alpha) + \alpha] + C.$$

- The economically relevant question is not just which schedules satisfy the differential equation, but which ones survive competition.
- The curve labeled JJ' starts too high at $\alpha = 0$. That lets bad projects sell to the market while retaining no equity, so it cannot be an equilibrium.
- The curve labeled LL' is feasible from an inference point of view, but it is dominated by KK' because entrepreneurs can signal the same μ with lower ownership cost under KK'.
- The schedule KK' is therefore the unique competitive equilibrium schedule.

Why it matters.

- This figure is the paper's visual proof that competition selects the least-cost separating schedule.

- It also explains why the equilibrium schedule must begin at the boundary where zero-retention entrepreneurs are exactly indifferent, not strictly better off.

5.2 Theorem I and Theorem II: monotonicity and the cost of signaling

Read:

- Theorem I says that higher insider retention reveals better project quality if and only if project demand is normal in an underlying portfolio problem.
- Theorem II says that signaling induces entrepreneurs to hold more of their own projects than under full information.

Why it matters.

- Theorem I is the core separation result.
- Theorem II is the core welfare result. It identifies exactly where the signaling cost shows up: in distorted portfolio holdings rather than in inefficient project selection.

5.3 Proposition I: efficient project selection

Read:

- A project is financed if and only if its true market value exceeds its cost.
- So signaling does not make the market fund too many bad projects or reject too many good projects at the extensive margin.

Why it matters.

- This is a subtle but important result. The model has asymmetric information and costly signaling, yet the acceptance set is still efficient.
- The distortion shows up in ownership structure, not in the yes/no investment decision.

5.4 Propositions II and III: specific risk, ownership, and welfare

Read:

- As specific risk Z rises, equilibrium insider ownership $\alpha^*(\mu)$ falls.
- At the same time, the entrepreneur's expected utility rises.

Why it matters.

- These results look surprising together until one remembers that the relevant object is *specific* risk, not total risk.
- A project that is more distinct from the market is easier to separate from other projects by insider ownership, so the signaling technology improves.

5.5 Proposition IV and the debt discussion

Read:

- Higher project variance implies lower optimal debt.
- Across otherwise similar firms, value and debt can be positively correlated because both reflect the underlying signal α.

Why it matters.

- The paper shows how one can observe a positive value-debt relationship without rejecting the causal content of Modigliani-Miller.
- Information can create observational correlations in capital structure that are absent in a symmetric-information world.

5.6 Financial intermediation: the big conceptual extension

Read:

- The intermediary solves an appropriability problem by embedding information in a portfolio.
- It solves a credibility problem by putting insider capital at risk.
- High leverage of intermediaries can emerge naturally when screened assets have low specific risk.

Why it matters.

- This is the paper's bridge from corporate-finance signaling to a theory of banking and intermediation.
- Even though the section is preliminary, it is one of the paper's most influential ideas.

6 Short summary

- **Method.** The paper builds a signaling model in which entrepreneurs know project quality and use insider ownership to transmit information to competitive capital markets.
- **Main conceptual result.** Financial structure carries information. In equilibrium, better projects are associated with higher insider retention and higher firm value.
- **Main efficiency result.** Signaling is costly because entrepreneurs overinvest in their own firms relative to the full-information optimum, but the set of projects financed is still efficient.
- **Capital-structure lesson.** Debt and value can be statistically related in cross section because both are tied to the informative ownership choice, even without taxes or bankruptcy costs.
- **Intermediation lesson.** Financial intermediaries can be understood as organizations that acquire information, hold assets on their own balance sheets, and signal information quality through insider stakes and leverage.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that asymmetric information can make insider ownership a credible signal of project quality.
- It also shows that once financial structure becomes informative, firm value and observed financing decisions need not be unrelated, even in a world without taxes.
- Finally, it argues that financial intermediation is a natural institutional response to informational asymmetry.

7.2 Economic intuition

- Good entrepreneurs are willing to keep more skin in the game because they know their projects are worth it.
- Bad entrepreneurs do not want to imitate them if imitation requires bearing too much undiversified risk.
- Markets therefore learn from ownership structure, and financing choices become part of the information system of the economy.

7.3 Takeaway

This paper is important because it links three ideas that are often studied separately: adverse selection, capital structure, and intermediation. Its central lesson is that ownership is informative. When insiders know more than markets do, the fraction they keep is a priced signal, and once that is true, financial structure cannot be analyzed as if it were merely a passive veil over real projects.